

Problem

For each of the following cases, find the complex power, the average power, and the reactive power:

(a) $v(t) = 169.7 \sin(377t + 45^\circ) \text{ V}$,
 $i(t) = 5.657 \sin(377t) \text{ A}$

(b) $v(t) = 339.4 \sin(377t + 90^\circ) \text{ V}$,
 $i(t) = 5.657 \sin(377t + 45^\circ) \text{ A}$

(c) $\mathbf{V} = 900 \angle 90^\circ \text{ V rms}$, $\mathbf{Z} = 75 \angle 45^\circ \Omega$

(d) $\mathbf{I} = 100 \angle 60^\circ \text{ A rms}$, $\mathbf{Z} = 50 \angle 60^\circ \Omega$

Step-by-step solution

Step 1 of 12

(a)

The value of the voltage $v(t)$ is,

$$v(t) = 169.7 \sin(377t + 45^\circ) \text{ V}$$

The value of the voltage in phasor form is,

$$\mathbf{V} = 169.7 \angle 45^\circ \text{ V}$$

The value of the current $i(t)$ is,

$$i(t) = 5.657 \sin(377t) \text{ A}$$

The value of the current in phasor form is,

$$\mathbf{I} = 5.657 \angle 0^\circ \text{ A}$$

Step 2 of 12

Determine the value of complex power $\mathbf{S}$.

$$\mathbf{S} = \frac{1}{2} \mathbf{V} \mathbf{I}^*$$

Substitute $5.657 \angle 0^\circ \text{ A}$ for $\mathbf{I}$ and $169.7 \angle 45^\circ \text{ V}$ for $\mathbf{V}$ in the equation.

$$\begin{aligned} \mathbf{S} &= \frac{1}{2} (169.7 \angle 45^\circ) (5.657 \angle 0^\circ)^* \\ &= (0.5) (169.7 \angle 45^\circ) (5.657 \angle 0^\circ) \\ &= 339.4 + j339.4 \text{ VA} \end{aligned}$$

Thus, the value of the complex power, $\mathbf{S}$ is $\boxed{(339.4 + j339.4) \text{ VA}}$.

Step 3 of 12

Compare $(339.4 + j339.4) \text{ VA}$ with general form of complex power $P + jQ$.

Here, P is the average or real power and Q is the reactive power.

$$P = 339.4 \text{ W}$$

$$Q = 339.4 \text{ VAR}$$

Thus, the value of average power, P is 339.4 W .

Thus, the value of reactive power, Q is 339.4 VAR .

Step 4 of 12

(b)

The value of the voltage $v(t)$ is,

$$v(t) = 339.4 \sin(377t + 90^\circ) \text{ V}$$

The value of the voltage in phasor form is,

$$V = 339.4 \angle 90^\circ \text{ V}$$

The value of the current $i(t)$ is,

$$i(t) = 5.657 \sin(377t + 45^\circ) \text{ A}$$

The value of the current in phasor form is,

$$I = 5.657 \angle 45^\circ \text{ A}$$

Step 5 of 12

Determine the value of complex power S .

$$S = \frac{1}{2} VI^*$$

Substitute $5.657 \angle 45^\circ \text{ A}$ for I and $339.4 \angle 90^\circ \text{ V}$ for V in the equation.

$$\begin{aligned} S &= \frac{1}{2} (339.4 \angle 90^\circ) (5.657 \angle 45^\circ)^* \\ &= (0.5) (339.4 \angle 90^\circ) (5.657 \angle -45^\circ) \\ &= 678.82 + j678.82 \text{ VA} \end{aligned}$$

Thus, the value of the complex power, S is $(678.82 + j678.82) \text{ VA}$.

Step 6 of 12

Compare $(678.82 + j678.82) \text{ VA}$ with general form of complex power $P + jQ$.

Here, P is the average or real power and Q is the reactive power.

$$P = 678.82 \text{ W}$$

$$Q = 678.82 \text{ VAR}$$

Thus, the value of average power, P is 678.82 W .

Thus, the value of reactive power, Q is 678.82 VAR .

Step 7 of 12

(c)

Determine the value of the current I .

$$I = \frac{V}{Z}$$

Substitute $900\angle 90^\circ \text{ V rms}$ for V and $75\angle 45^\circ \Omega$ for Z in the equation.

$$\begin{aligned} I &= \frac{900\angle 90^\circ \text{ V rms}}{75\angle 45^\circ \Omega} \\ &= 12\angle 45^\circ \text{ A rms} \end{aligned}$$

Step 8 of 12

Determine the value of complex power S .

$$S = VI^*$$

Substitute $12\angle 45^\circ \text{ A rms}$ for I and $900\angle 90^\circ \text{ V rms}$ for V in the equation.

$$\begin{aligned} S &= (900\angle 90^\circ)(12\angle 45^\circ)^* \\ &= (900\angle 90^\circ)(12\angle -45^\circ) \\ &= (7.637 + j7.637) \text{ kVA} \end{aligned}$$

Thus, the value of the complex power, S is $(7.637 + j7.637) \text{ kVA}$.

Step 9 of 12

Compare $(7.637 + j7.637) \text{ kVA}$ with general form of complex power $P + jQ$.

Here, P is the average or real power and Q is the reactive power.

$$P = 7.637 \text{ kW}$$

$$Q = 7.637 \text{ kVAR}$$

Thus, the value of average power, P is 7.637 kW .

Thus, the value of reactive power, Q is 7.637 kVAR .

Step 10 of 12

(d)

Determine the value of the voltage V .

$$V = IZ$$

Substitute $100\angle 60^\circ \text{ A rms}$ for I and $50\angle 60^\circ \Omega$ for Z in the equation.

$$\begin{aligned} V &= (100\angle 60^\circ)(50\angle 60^\circ) \\ &= 5000\angle 120^\circ \text{ V rms} \end{aligned}$$

Step 11 of 12

Determine the value of complex power S .

$$S = VI^*$$

Substitute $100\angle 60^\circ$ A rms for I and $5000\angle 120^\circ$ V rms for V in the equation.

$$\begin{aligned} S &= (5000\angle 120^\circ)(100\angle 60^\circ)^* \\ &= (5000\angle 120^\circ)(100\angle -60^\circ) \\ &= (250 + j433) \text{ kVA} \end{aligned}$$

Thus, the value of the complex power, S is $(250 + j433) \text{ kVA}$.

Step 12 of 12

Compare $(250 + j433) \text{ kVA}$ with general form of complex power $P + jQ$.

Here, P is the average or real power and Q is the reactive power.

$$P = 250 \text{ kW}$$

$$Q = 433 \text{ kVAR}$$

Thus, the value of average power, P is 250 kW .

Thus, the value of reactive power, Q is 433 kVAR .